

Test of Hypothesis



Statistical Hypothesis

- **Hypothesis** is a statement about one or more populations.
- **Statistical hypotheses** are hypotheses that are stated in such a way that they may be evaluated by appropriate statistical techniques.

Types of Statistical Hypotheses

- The **null hypothesis** is a hypothesis which states that there is no difference between assumed and actual value of the parameter. It is usually denoted by H_0 .

As for example, In inoculation and cholera data,

H_0 : Inoculation has no effect on Cholera.

- The **alternative hypothesis** is a statement of what we will believe is true if our sample data cause us to reject the null hypothesis. It is usually denoted H_1 .

In other words, the alternative hypothesis is that one which the researcher wishes to support, while the null hypothesis, a contradiction to the alternative hypothesis.

For example- if we want to test the null hypothesis ($H_0: \mu = \mu_0$) then the H_1 could be:



$H_1: \mu \neq \mu_0$	Two-tailed alternatives
$H_1: \mu > \mu_0$	right-tailed alternatives
$H_1: \mu < \mu_0$	left-tailed alternatives

Types of Errors in Hypotheses tests

- There could be two types of errors in a hypothesis testing procedure. They are named as **Type I** and **Type II error**.
- The **four** possible situations that arise in any test procedure are given below:

1. Two-tailed test ($\neq$)

Use when you only want to check whether there is any difference, without specifying the direction.

Example: A company claims the mean battery life is 10 hours. You want to test if the actual mean is different from 10 hours (could be more or less).

$$H_0: \mu = 10$$

$$H_1: \mu \neq 10$$

2. Right-tailed test ($>$)

Use when you want to test if the parameter is greater than a certain value.

Example: A new fertilizer is claimed to increase crop yield beyond 50 kg/acre.

$$H_0: \mu \leq 50$$

$$H_1: \mu > 50$$

3. Left-tailed test (<)

Use when you want to test if the parameter is **less** than a certain value.

Example:

A manufacturer claims tar content of cigarettes is **at most 17.5 mg**.

You suspect the average tar content is **less**.

$$H_0: \mu \geq 17.5$$

$$H_1: \mu < 17.5$$

Types of Errors in Hypotheses tests

Decision	State of nature	
	H_0 is True	H_0 is False
Accept H_0	Correct decision (No error)	Type II error
Reject H_0	Type I error	Correct decision (No error)

Level of significance, $\alpha = P(\text{type I error}) = P(\text{reject } H_0 \mid H_0 \text{ is true})$

Level of significance

- ❑ The **level of significance** (often denoted by α) is a threshold we set in hypothesis testing to decide whether to reject the null hypothesis (H_0).
- ❑ It represents the probability of making a **Type I error**—rejecting the null hypothesis when it is actually true.

👉 Example:

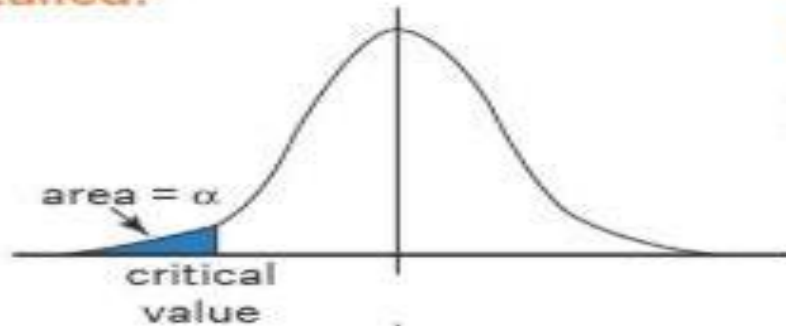
If $\alpha = 0.05$, it means we are willing to accept a **5% risk** of concluding that “something is happening” when in truth **nothing is happening**.

🔑 Key points:

- ❑ Common values: **0.05 (5%)**, **0.01 (1%)**, or **0.10 (10%)**.
- ❑ If the **p-value** $\leq \alpha$, we reject the null hypothesis.
- ❑ If the **p-value** $> \alpha$, we fail to reject the null hypothesis.

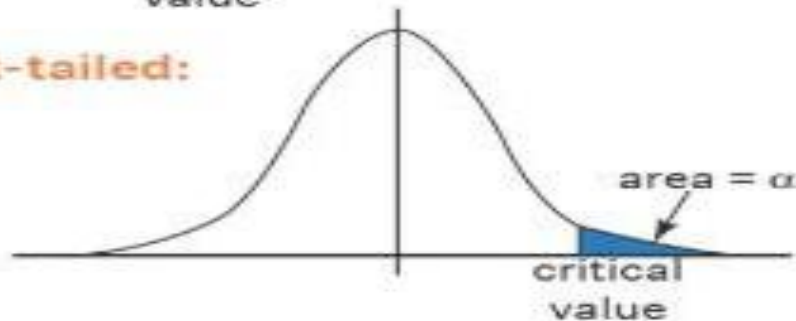
Critical value: The dividing point between the region where the null hypothesis is rejected and the region where it is not rejected

left-tailed:

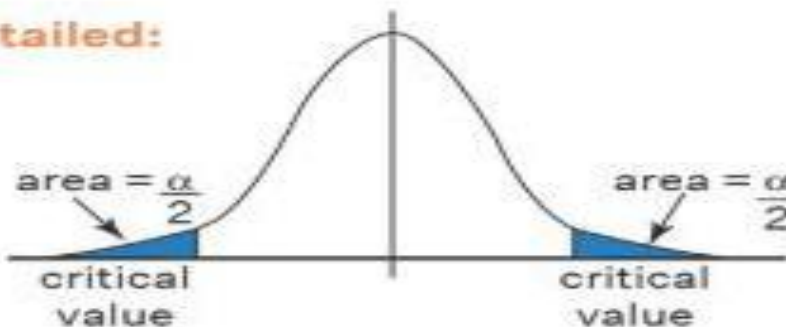


■ - Reject H_0
□ - Do not reject H_0

right-tailed:



two-tailed:



Test statistic

- A statistic on which the decision whether to accept or reject H_0 is based is called a **test statistic**.

How to perform a test

The following steps are followed in a complete test of hypothesis procedure:

- Step 1: Assumptions / situation
- Step 2: Setting up the hypothesis
- Step 3: The significance level
- Step 4: The test statistic
- Step 5: The Decision making

Known test statistics and their sampling distribution

The Z-statistic: One sample mean situation

$$Z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$$

Here, $H_0 : \mu = \mu_0$

Assumptions:

- If $X \sim N(\mu, \sigma^2)$, $Z \sim N(0, 1)$
- In each of the above cases σ is required to be known. If σ is not known the distribution of z is NOT $N(0,1)$

Decision making

	Decision rule	
H1	$\alpha = 0.05$ Reject H_0 if and only if	$\alpha = 0.01$ Reject H_0 if and only if
$\mu > \mu_0$	$Z \geq 1.645$	$Z \geq 2.33$
$\mu < \mu_0$	$Z \leq -1.645$	$Z \leq -2.33$
$\mu \neq \mu_0$	$Z \geq 1.96$ Or $Z \leq -1.96$	$Z \geq 2.58$ Or $Z \leq -2.58$

Illustration-1

A bulb manufacturing company claims that the **average** longevity of their bulb is 4 years with a **standard deviation** of 0.16 years. A **random sample** of 40 bulbs gave a **mean** longevity of 3.45 years. Test the hypothesis that the mean longevity of the bulbs produced by the company is 4 years. Use a 5 percent level of significance.

Solution: Here $H_0: \mu = 4$; $H_1 : \mu \neq 4$

At 5% significance, $Z_{\alpha/2} = \pm 1.96$

Here, $\bar{x} = 3.45$, $n = 40$, $\mu_0 = 4$ and $\sigma = 0.16$

$$Z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}} = -21.7$$

Since $Z \leq -1.96$, hence we reject the null hypothesis and conclude that the mean longevity of the bulb is not 4 years.

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- **H.W.:** The **average IQ** of university women in Bangladesh is suspected to be **more than** the average 110 for all students. A random **sample** of 64 women yielded an average IQ of 115.5 and a standard deviation of 20. Can you conclude that the average score of the women in the population is really **more than** 110? Test this at 5 percent level of significance.

The Z-statistic: Two sample mean situation

Example: A buyer of electric bulbs bought 100 bulbs each of two famous brands. Upon testing these, he found that brand A had a mean life of 1500 hours with a standard deviation of 50 hours whereas brand B had a mean life of 1530 hours with a standard deviation of 60 hours. Can it be concluded at 5 percent level of significance that the two brands differ significantly in quality of the bulbs.

Given:

- Mean life of brand A:
 $\bar{x}_1 = 1500, s_1 = 50, n_1 = 100$
- Mean life of brand B:
 $\bar{x}_2 = 1530, s_2 = 60, n_2 = 100$
- Significance level: $\alpha = 0.05$

Hypotheses:

- Null Hypothesis $H_0: \mu_1 = \mu_2$ (no difference in mean life)
- Alternative Hypothesis $H_1: \mu_1 \neq \mu_2$ (there is a difference)

This is a two-tailed test.

Test Statistic (Z):

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Substitute the values:

$$Z = \frac{1500 - 1530}{\sqrt{\frac{50^2}{100} + \frac{60^2}{100}}} = \frac{-30}{\sqrt{25 + 36}} = \frac{-30}{\sqrt{61}} \approx \frac{-30}{7.81} \approx -3.84$$

Critical Value at 5% level (two-tailed):

From the Z-table, the critical values for $\alpha = 0.05$ (two-tailed) are:

$$Z_{0.025} = \pm 1.96$$

Decision:

$$|Z| = 3.84 > 1.96 \Rightarrow \text{Reject } H_0$$

Conclusion:

At the 5% level of significance, there is **sufficient evidence** to conclude that the **two brands of bulbs differ significantly** in quality (mean life).

The t-statistic: One sample mean situation

$$t = \frac{\bar{x} - \mu_0}{s / \sqrt{n}}$$

Following statements can be made on sampling distribution of t under $H_0 : \mu = \mu_0$

- In case of using t-statistic, σ is required to be estimated from the sample (i.e. σ is not known). If σ was known, we could use the z-statistic
- If $X \sim N(\mu, \sigma^2)$, $t \sim t_{n-1}$

Decision making

	Decision rule	
H_1	Rejection region (Reject H_0 if and only if)	Acceptance region
$H_1: \mu > \mu_o$	$t_{cal} \geq t_{\alpha, (n-1)}$	$t_{cal} < t_{\alpha, (n-1)}$
$H_1: \mu < \mu_o$	$t_{cal} \leq -t_{\alpha, (n-1)}$	$t_{cal} > -t_{\alpha, (n-1)}$
$H_1: \mu \neq \mu_o$	$ t_{cal} \geq t_{\alpha/2, (n-1)}$	$ t_{cal} < t_{\alpha/2, (n-1)}$

Confidence Interval

$$P\{-t_{\alpha/2, (n-1)} \leq t \leq t_{\alpha/2, (n-1)}\} = 1 - \alpha$$

$$\Rightarrow P\{-t_{\alpha/2, (n-1)} \leq \mathbf{t = \frac{\bar{x} - \mu}{s / \sqrt{n}}} \leq t_{\alpha/2, (n-1)}\} = 1 - \alpha$$

$$\Rightarrow P\{-(s / \sqrt{n}) t_{\alpha/2, (n-1)} \leq \bar{x} - \mu \leq (s / \sqrt{n}) t_{\alpha/2, (n-1)}\} = 1 - \alpha$$

$$\Rightarrow P\{(s / \sqrt{n}) t_{\alpha/2, (n-1)} \geq -\bar{x} + \mu \geq -(s / \sqrt{n}) t_{\alpha/2, (n-1)}\} = 1 - \alpha$$

$$\Rightarrow P\{\bar{x} - (s / \sqrt{n}) t_{\alpha/2, (n-1)} \leq \mu \leq \bar{x} + (s / \sqrt{n}) t_{\alpha/2, (n-1)}\} = 1 - \alpha$$

Confidence Interval for μ

$$P\{\bar{x} - (s / \sqrt{n}) t_{\alpha/2, (n-1)} \leq \mu \leq \bar{x} + (s / \sqrt{n}) t_{\alpha/2, (n-1)}\} = 1 - \alpha$$

$$\Rightarrow P\{64.68 \leq \mu \leq 68.12\} = 0.95$$

95% CI for μ is (64.68, 68.12)

Example

- A random sample of 8 cigarettes of certain brand has an average tar content of 18.6 mg and standard deviation of 2.4 mg. The manufacturer, however, claims the average content does not exceed 17.5 mg. Use 0.01 level of significance to test the manufacturer's claim.

P-value

1. The Null Hypothesis (H_0)

H_0 is our “default assumption” — for example:

$$H_0: \mu = 120.$$

2. Check how unusual the sample is

- If the sample result is **very far (extreme)** from what H_0 predicts, it is considered **unlikely under H_0** . A result **very close to 120** is **not extreme**. A result **far from 120** (say, 135 or 105) is **more extreme**.
- Suppose we observed a sample mean of 135. That’s quite far from 120.
- The p-value asks: If μ really = 120, what’s the probability of getting 135 or something even further away (like 140, 145, ... OR in the other direction, 105, 100, ...)?
- The **p-value** tells us exactly how unlikely it is.

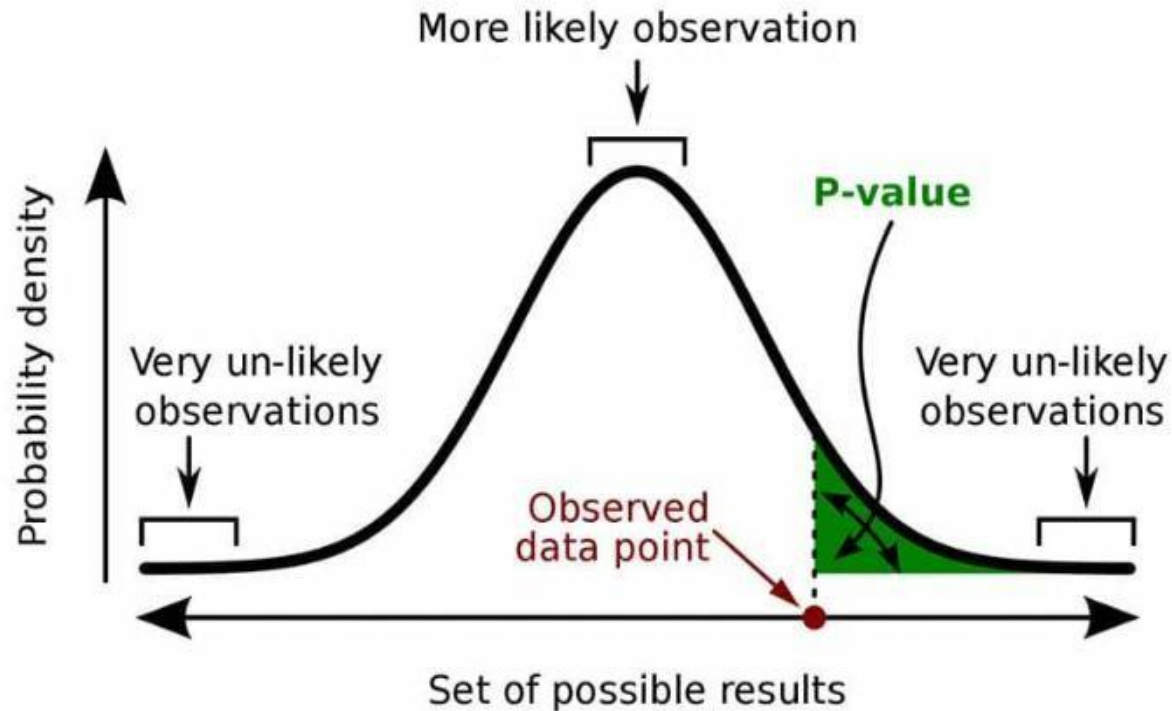
3. Decision rule

- If the p-value is small (smaller than our significance level, e.g., 0.05), it means:
- “This data would almost never happen if H_0 were true.”
- Since it’s so unlikely, it suggests that H_0 is not a good explanation for the data, or H_0 doesn’t explain the data well.

4. Conclusion

- Because the data is incompatible with H_0 , we reject H_0 .
- If the p-value were large, the data is consistent with H_0 , so we fail to reject H_0 .

P-value



A **p-value** (shaded green area) is the probability of an observed (or more extreme) result assuming that the null hypothesis is true.

Understanding the p-Value

- A **p-value** tells us how likely our data would be *if the null hypothesis were actually true*. It's basically a way to judge how strongly the data speak **against** the null.
- The **smaller** the p-value, the **stronger** the evidence to reject the null hypothesis.

➤ Conclusions Using p -Values

If $p\text{-value} \leq \alpha$, then reject the null hypothesis.

If $p\text{-value} > \alpha$, then fail to reject the null hypothesis.

What does a p-value of 0.07 mean?

- A **p-value** tells you how strong the evidence is against the null hypothesis.
- A **p = 0.07** means: **7% chance** of getting the observed result (or something more extreme) *if the null hypothesis is true*.

➤ What does “significant at 10% but not at 5%” mean?

- We compare the p-value with a **significance level (α)**.
- Common significance levels are: **10% (0.10), 5% (0.05), 1% (0.01)**

Now compare $p = 0.07$:

- **✓ $0.07 < 0.10 \rightarrow$ Significant at 10% level**

(meaning: result is considered significant if we allow a 10% chance of error)

- **✗ $0.07 > 0.05 \rightarrow$ NOT significant at 5% level**

(meaning: if we require stronger evidence—allow only 5% chance of error—it's not strong enough)

Test hypothesis using p-value

A manufacturer of chocolate chips would like to know whether its bag filling machine is underfilling the bags compared to the 409-gram setting. It is believed that the machine is underfilling the bags. A 40 bag sample had a mean of 402.0 grams. A level of significance of 0.02 will be used. Assume the standard deviation is known to be 15.0 grams.

- (a) State the hypotheses.
- (b) Compute the test statistic.
- (c) Make a decision to reject or fail to reject the null hypothesis based on the p -value.
- (d) Make a decision to reject or fail to reject the null hypothesis based on critical value.

Example-Answer

Answer:

(a)

$$H_0: \mu = 409$$

$$H_A: \mu < 409$$

(b) Here $n = 40$, $\mu = 409$, $\bar{x} = 402$, $\sigma = 15$, $\alpha = 0.02$.

Test statistic is:

$$z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{402 - 409}{\frac{15}{\sqrt{40}}} = -2.95$$

(c) **p-value approach:**

The p -value = $P(Z < -2.95) = 0.0016$ $z = -2.95$

(can be find from a standard normal table, z-table).

Decision/ Comment: At $\alpha = 0.02$, $p\text{-value} \leq \alpha$, then we may reject the null hypothesis. That is, the bag filling machine is underfilling the bags.

Example-Cont..

(d) **Critical value approach:**

The rejection region is $Z \leq -Z_{\alpha}$; where, $Z_{\alpha} = Z_{0.02} = 2.054$.

Decision/ Comment: At $\alpha = 0.02$, $Z \leq -Z_{\alpha}$, then we may reject the null hypothesis. That is, the bag filling machine is underfilling the bags.

Example 2. A manufacturer of chocolate chips would like to know whether its bag filling machine is overfilling the bags compared to the 409-gram setting. It is believed that the machine is underfilling the bags. A 40 bag sample had a mean of 402.0 grams. A level of significance of 0.02 will be used. Assume the standard deviation is known to be 15.0 grams.

(Hints. The p -value = $P(Z > -2.95)$
 $= 1 - P(Z \leq -2.95) = 1 - 0.0016 = 0.9984$)

Testing for a Population Proportion

The test statistic for a hypothesis test for a population mean is

$$Z = \frac{\hat{p} - p}{\sqrt{\frac{p(1-p)}{n}}}$$

where p is the population proportion and $\hat{p}$ is the sample proportion. If the null hypothesis $H_0: p = p_0$ is true then the test statistic becomes

$$Z = \frac{\hat{p} - p}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

Example 1.

A company is considering launching a new product in the market. A market research firm surveyed a sample of 1500 consumers and found that 47% of them are interested in purchasing the product. Using this data, the company's marketing team wants to test the claim that the percentage of consumers interested in the product is above 44% at the 0.1 significance level.

- (a) State the hypotheses.
- (b) Compute the test statistic.
- (c) Make a decision to reject or fail to reject the
- (d) null hypothesis based on the p-value.
- (e) Make a decision to reject or fail to reject the
- (f) null hypothesis based on critical value.

Example 1.

(a) $H_0 : p = 0.44$
 $H_A : p > 0.44$

(b) Here $n = 1500$, $p = 0.44$, $\hat{p} = 0.47$, $\alpha = 0.1$ and $z_\alpha = z_{0.10} = 1.28$.

Test statistic is $Z = \frac{0.47 - 0.44}{\sqrt{\frac{0.44(1-0.44)}{1500}}} = 2.34$

(c) p-value approach: The p -value is, $p = 0.0096$.
Decision: Since $p\text{-value} \leq \alpha$, we reject the null hypothesis.
That is, the percentage of customers who are interested in purchasing the product is above 44%.

(d) Critical value approach: The rejection region is $Z \geq Z_\alpha$; where $Z_\alpha = Z_{0.10} = 1.28$.
Decision: At $\alpha = 0.10$, $Z \geq Z_\alpha$, then we may reject the null hypothesis. That is, the percentage of customers who are interested in purchasing the product is above 44%.

Example 2.

A publisher reports that 63% of their readers own a personal computer. A marketing executive wants to test the claim that the percentage is actually different from the reported percentage. A random sample of 110 found that 56% of the readers owned a personal computer. Answer the following questions at the 0.05 level.

- (a) State the hypotheses.
- (b) Compute the test statistic.
- (c) Compute the p -value.
- (d) Make a decision to reject or fail to reject the null hypothesis.
- (e) Make a decision.

Example 2.

(a)

$$H_o: p = 0.63$$

$$H_A: p \neq 0.63$$

(b) Here $n = 110$, $p = 0.63$, $p^* = 0.56$, $\alpha = 0.05$.

$$\text{Test statistic is } Z = \frac{0.56 - 0.63}{\sqrt{\frac{0.63(1-0.63)}{110}}} = -1.52$$

(c) The p -value is $2(0.0643) = 0.1286$. Since $p\text{-value} > \alpha$, we fail to reject the null hypothesis. We fail to reject the publisher's claim that 63% of the readers own a personal computer.

(d)

(e) The rejection region is $Z < -1.96$ or $Z > 1.96$. Hence, we fail to reject the null hypothesis.

Example 2.

(a)

$$H_0: p = 0.63$$

$$H_A: p \neq 0.63$$

(b) Here $n = 110$, $p = 0.63$, $\hat{p} = 0.56$, $\alpha = 0.05$.

$$\text{Test statistic is } Z = \frac{0.56 - 0.63}{\sqrt{\frac{0.63(1-0.63)}{110}}} = -1.52$$

(c) The p -value is $2(0.0643) = 0.1286$. Since $p\text{-value} > \alpha$, we fail to reject the null hypothesis. We fail to reject the publisher's claim that 63% of the readers own a personal computer.

(d)

(e) The rejection region is $Z < -1.96$ or $Z > 1.96$. Hence, we fail to reject the null hypothesis.